

Math 350 Section 1 and 2
Fall 2025
Exam II
October 22, 2025
Time Limit: 1 Hour 50 Minutes

Name (Print): _____

Student ID: _____

This exam contains 8 pages (including this cover page) and 7 problems. Check to see if any pages are missing. Enter all requested information on the top of this page, and put your initials on the top of every page, in case the pages become separated.

You may *not* use your books or notes on this exam.

You are required to show your work on each problem on this exam. The following rules apply:

- **Organize your work**, in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little credit.
- **Mysterious or unsupported answers will not receive full credit.** A correct answer, unsupported by calculations, explanation, or algebraic work will receive no credit; an incorrect answer supported by substantially correct calculations and explanations might still receive partial credit. This especially applies to limit calculations.
- If you need more space, use the back of the pages; clearly indicate when you have done this.
- **Box Your Answer** where appropriate, in order to clearly indicate what you consider the answer to the question to be.

Problem	Points	Score
1	10	
2	10	
3	10	
4	10	
5	10	
6	10	
7	10	
Total:	70	

Do not write in the table to the right.

1. (10 points) **TRUE or FALSE!** Write TRUE if the statement is true. Otherwise, write FALSE. Your response should be in ALL CAPS. No justification is required.

(a) a monotone increasing sequence must converge

(b) in any metric space a closed and bounded set must be compact

(c) if a set is not closed, then it must be open

(d) in a metric space with the discrete metric, every set is both open and closed

(e) if $\lim_{x \rightarrow 0} f(x) = 4$ then $f(0) = 4$

2. (10 points)

(a) Write down the definition of a metric space.

(b) Write down three different examples of metrics we can define on $\mathbb{R}^2$. (Not just the names!)

(c) Explain why $d(x, y) = (x - y)^2$ is NOT a metric on $\mathbb{R}$.

3. (10 points)

- (a) Write down the definition of a sequence $\{x_n\}$ of values in a metric space (M, d) converging to a value $L \in M$.
- (b) Prove that if a sequence $\{x_n\}$ converges to a value $L \in M$, then there exists a constant $R > 0$ such that $d(x_n, L) < R$ for all $n \in \mathbb{Z}_+$ without using any theorems from class.

4. (10 points)

(a) Write down the definition of a sequence $\{x_n\}$ in a metric space (M, d) being Cauchy

(b) Give an example of a metric space with a Cauchy sequence that does not converge.

(c) Suppose that $M = \mathbb{R}$ and d is the discrete metric. Prove that if $\{x_n\}$ is a Cauchy sequence, then the set

$$X = \{x_1, x_2, x_3, \dots\}$$

is finite.

5. (10 points) Carefully state each of the following theorems.

(a) the Lindelöf Covering Theorem

(b) the Bolzano-Weierstrass Theorem

(c) the Cantor Intersection Theorem

(d) the Heine-Borel Theorem

6. (10 points)

- (a) Let (S, d_S) and (T, d_T) be metric spaces and $f : S \rightarrow T$ a function, and suppose $a \in S$. Write the definition of $\lim_{x \rightarrow a} f(x) = L$.

- (b) Consider the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} 1, & x \in \mathbb{Q} \\ 0, & x \notin \mathbb{Q} \end{cases}$$

Prove that the limit $\lim_{x \rightarrow 0} f(x)$ does not exist.

- (c) Consider the function $f : [0, \infty) \rightarrow \mathbb{R}$ defined by $f(x) = \sqrt{x}$. Prove that $\lim_{x \rightarrow 4} f(x) = 2$ using the definition of a limit.

7. (10 points)

(a) Write down the definition of a subset $A \subseteq M$ of a metric space M being compact.

(b) Prove that if $A = \{a\}$ for some $a \in M$, then A is compact.